DIAGRAM INDEX

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DriveIQ

Usage-Based Vehicle Insurance Platform

UML Diagram Suite

11	4	9	3
Diagrams	Actors	Entities	Workflows

PREPARED BY

Bhavana Sri Tejavath

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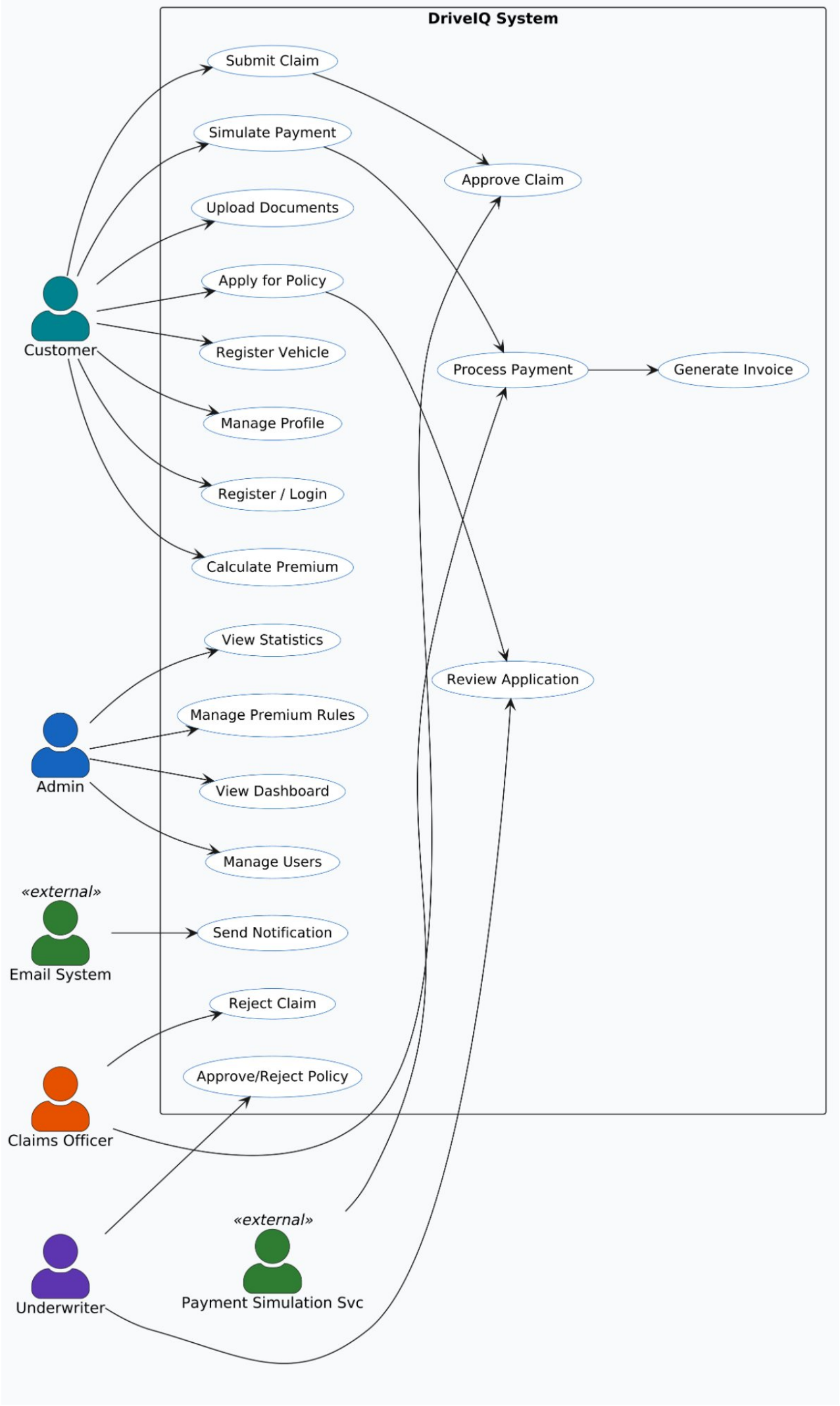
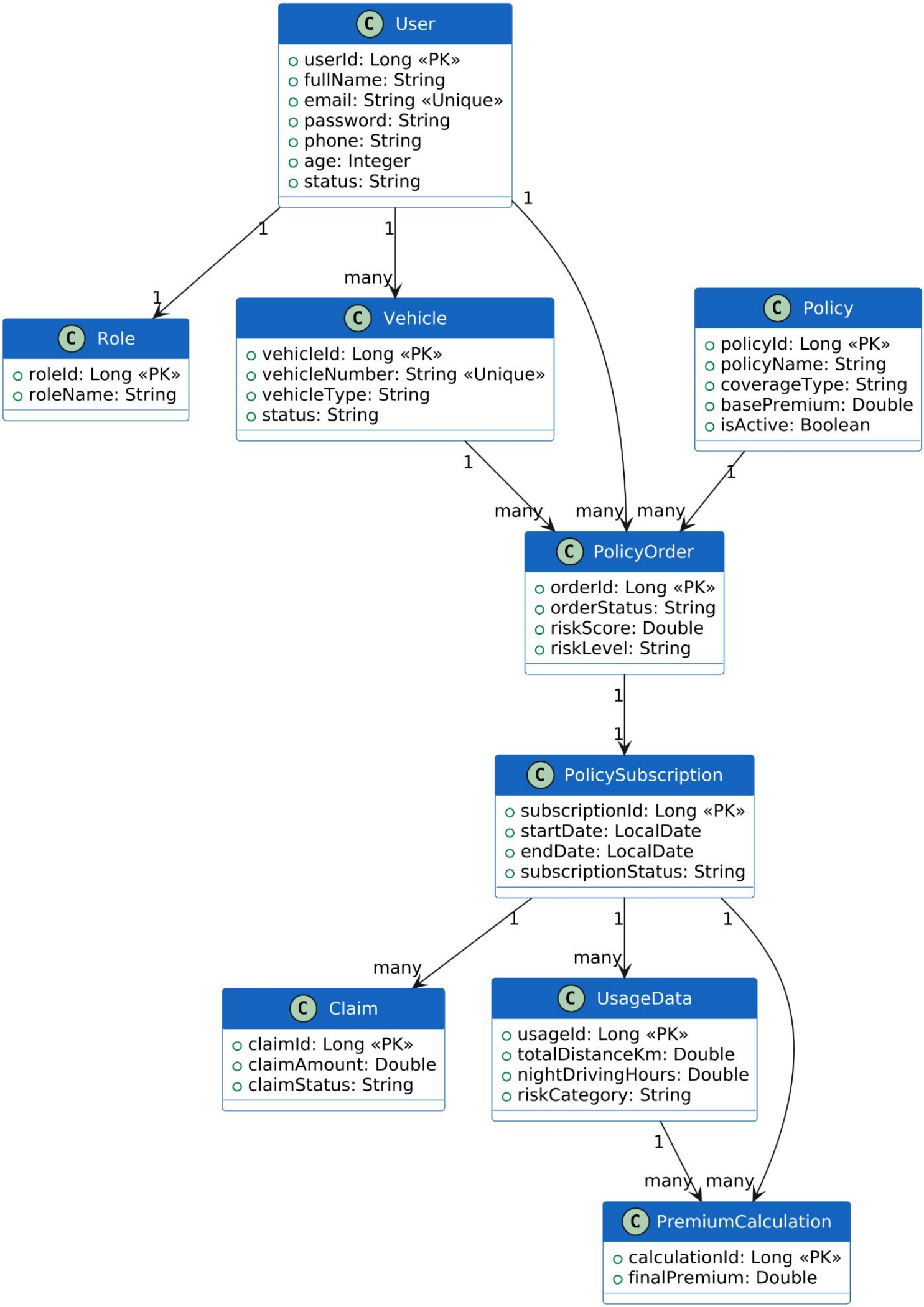
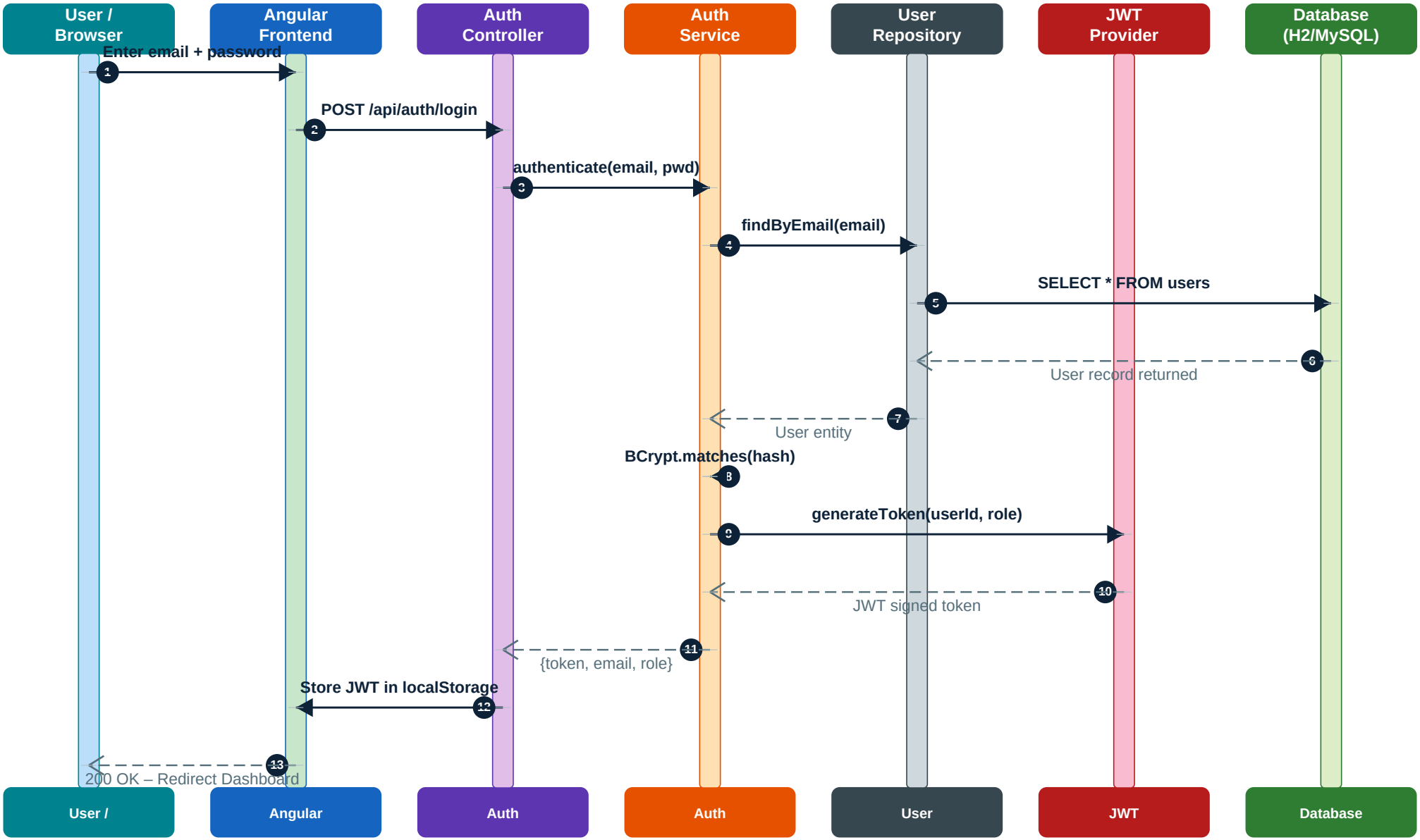
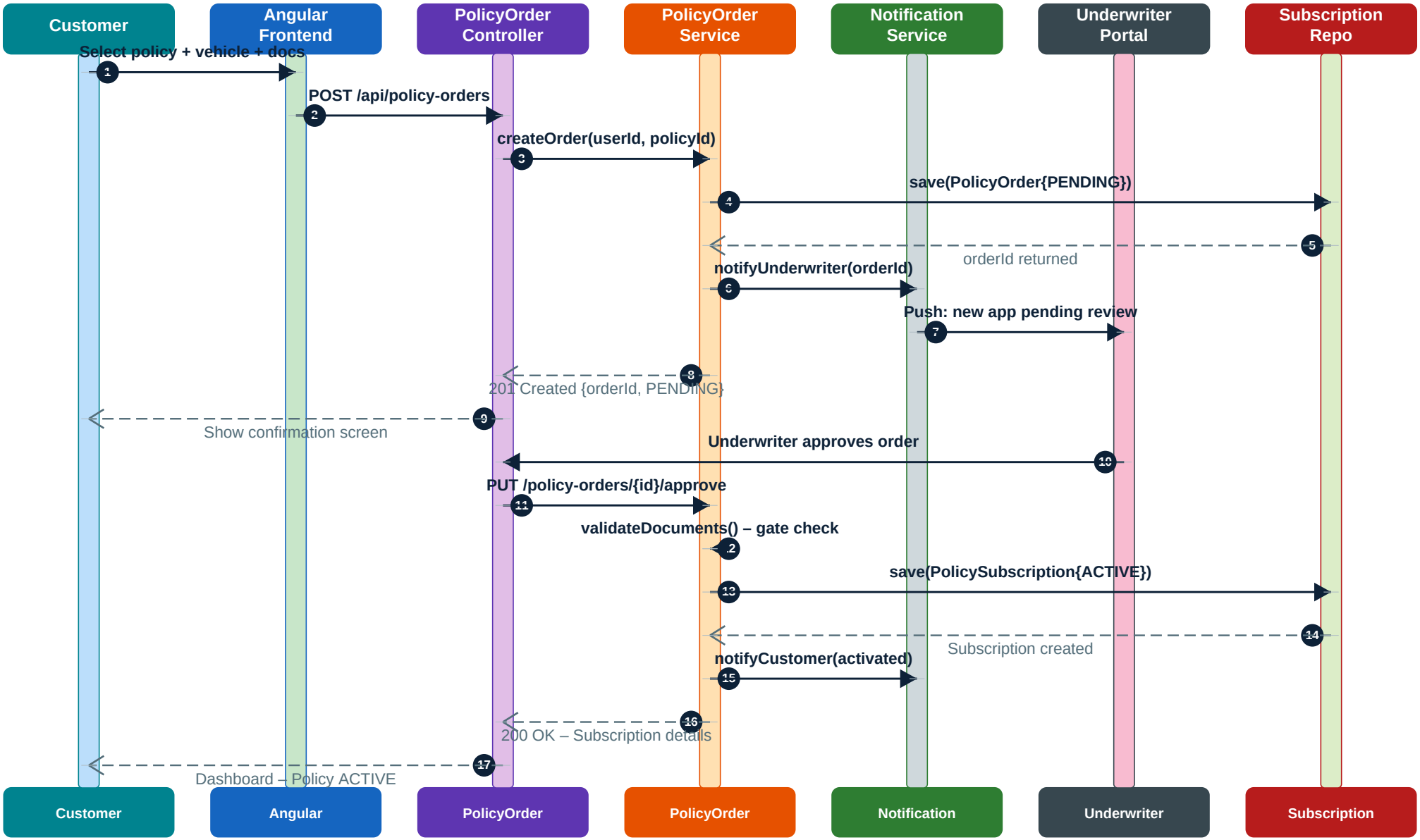
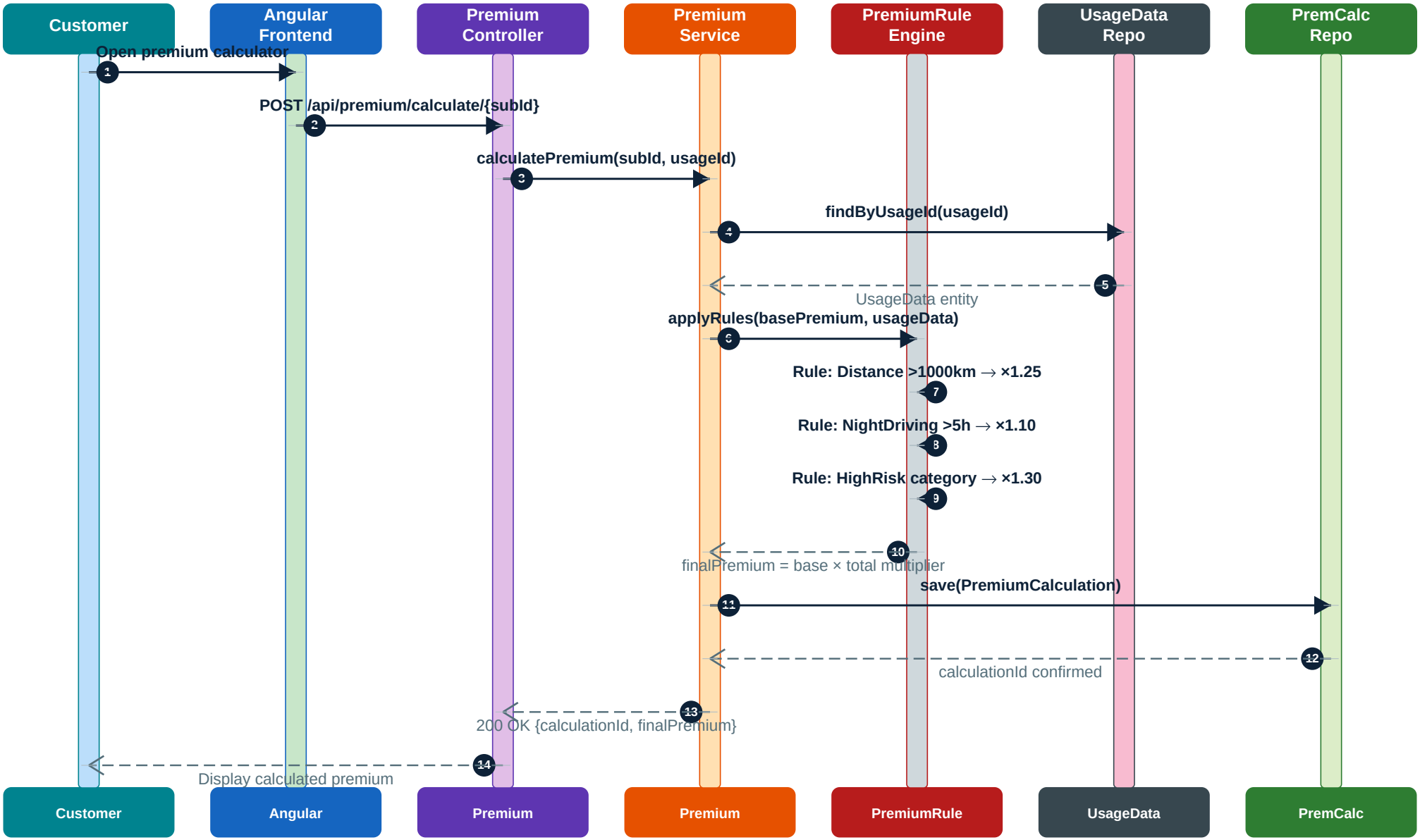
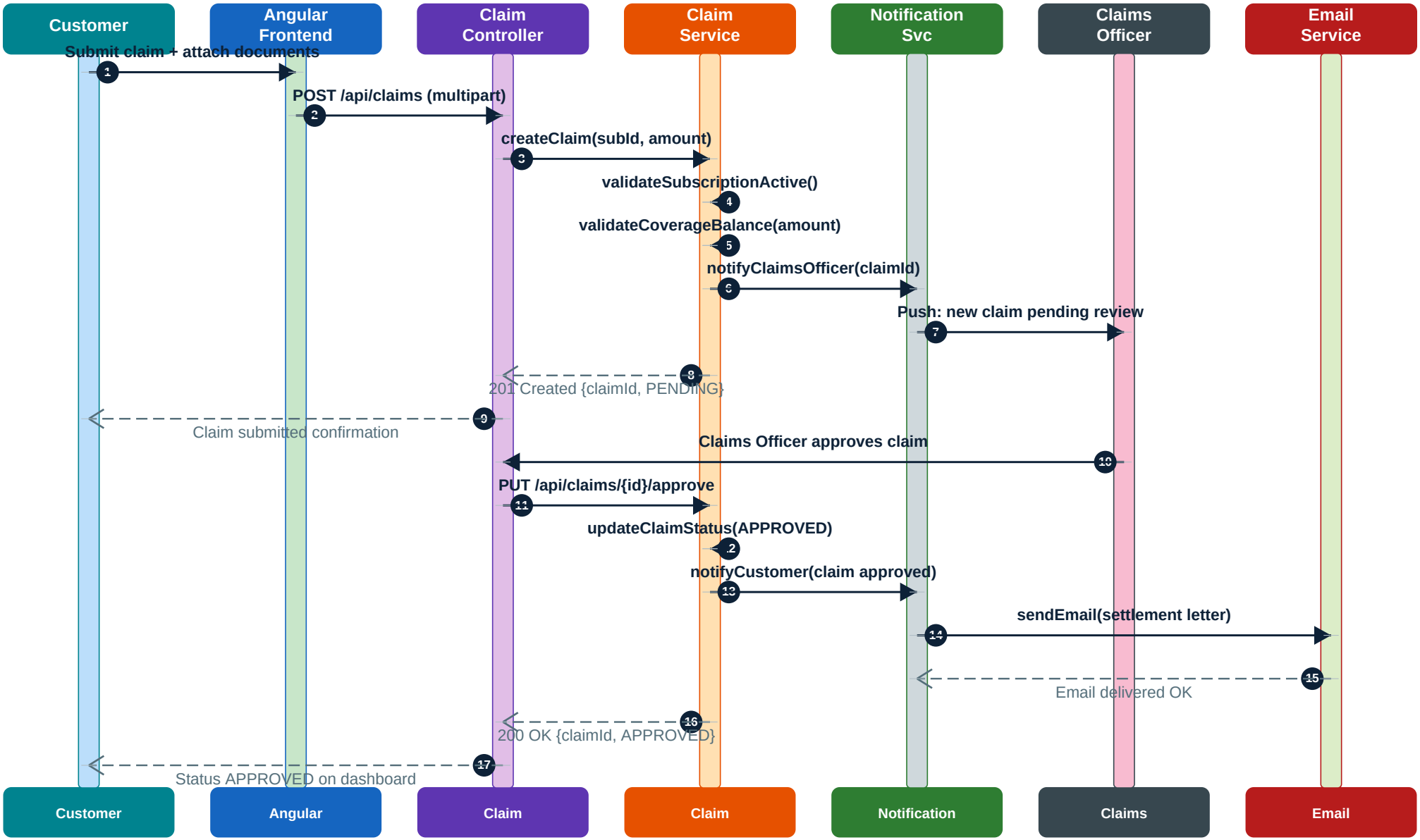

DriveIQ System
Submit Claim
Simulate Payment
Upload Documents
Apply for Policy
Register Vehicle
Manage Profile
Register / Login
Calculate Premium
View Statistics
Manage Premium Rules
View Dashboard
Manage Users
Send Notification
Reject Claim
Approve/Reject Policy
Approve Claim
Process Payment
Generate Invoice
Review Application
Customer
Admin
«external»
Email System
Claims Officer
Underwriter
«external»
Payment Simulation Svc

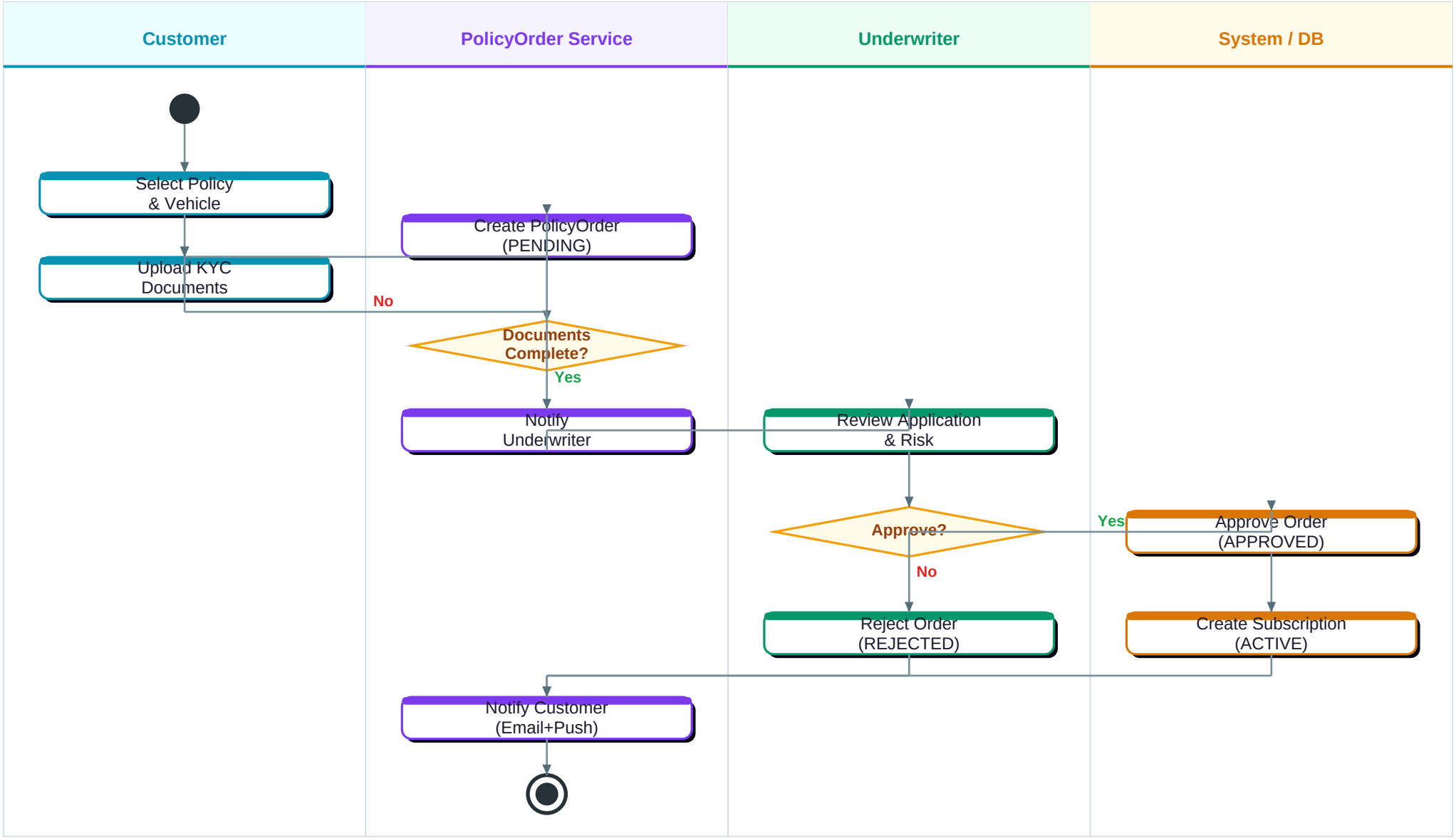

User
userId: Long «PK»
fullName: String
email: String «Unique»
password: String
phone: String
age: Integer
status: String
Role
roleId: Long «PK»
roleName: String
Vehicle
vehicleId: Long «PK»
vehicleNumber: String «Unique»
vehicleType: String
status: String
Policy
policyId: Long «PK»
policyName: String
coverageType: String
basePremium: Double
isActive: Boolean
PolicyOrder
orderId: Long «PK»
orderStatus: String
riskScore: Double
riskLevel: String
PolicySubscription
subscriptionId: Long «PK»
startDate: LocalDate
endDate: LocalDate
subscriptionStatus: String
Claim
claimId: Long «PK»
claimAmount: Double
claimStatus: String
UsageData
usageId: Long «PK»
totalDistanceKm: Double
nightDrivingHours: Double
riskCategory: String
PremiumCalculation
calculationId: Long «PK»
finalPremium: Double
1
many

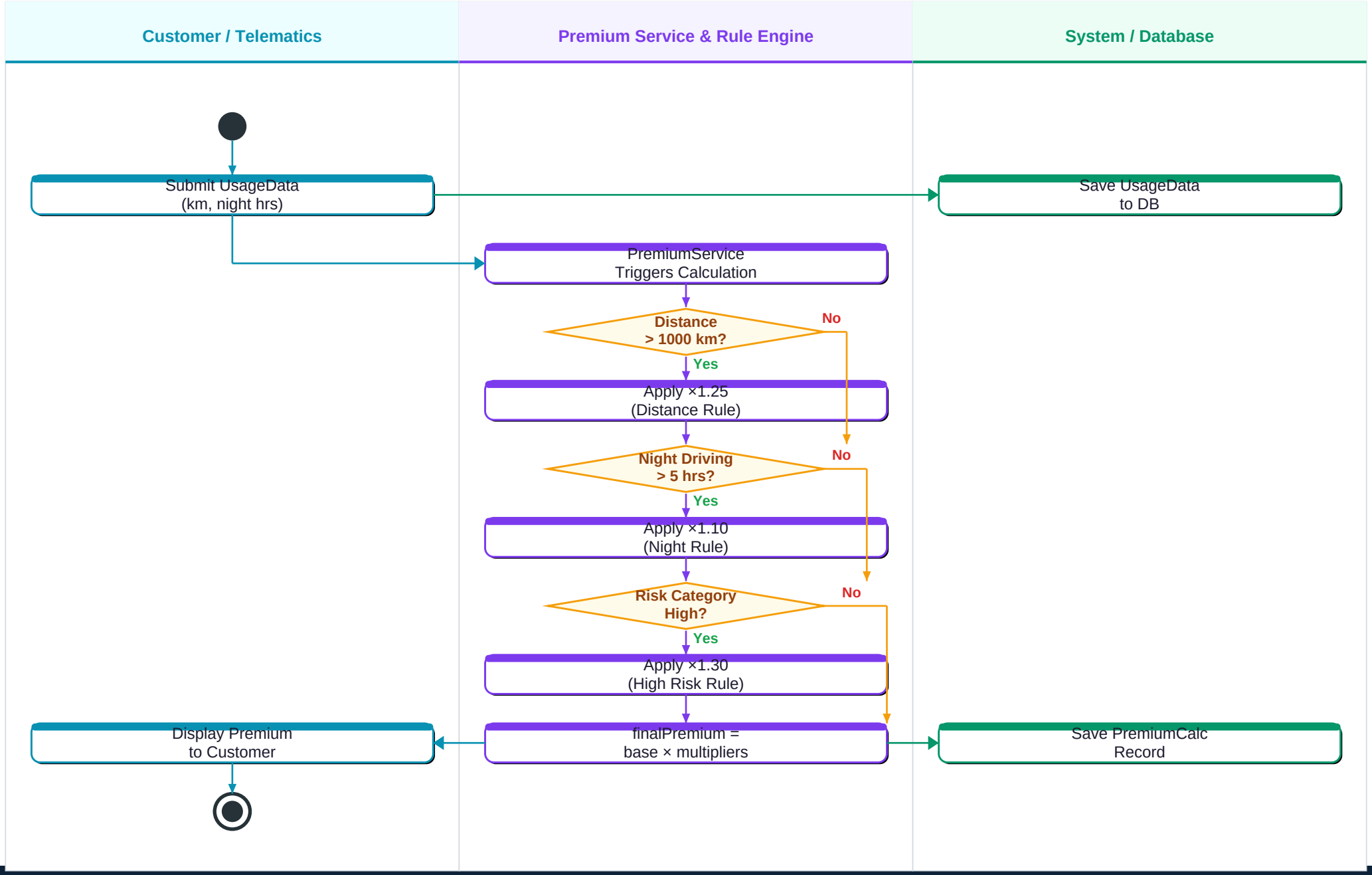

User / Browser
Angular Frontend
Auth Controller
Auth Service
User Repository
JWT Provider
Database (H2/MySQL)
1 Enter email + password
2 POST /api/auth/login
3 authenticate(email, pwd)
4 findByEmail(email)
5 SELECT * FROM users
6 User record returned
7 User entity
8 BCrypt.matches(hash)
9 generateToken(userId, role)
10 JWT signed token
11 {token, email, role}
12 Store JWT in localStorage
13 200 OK – Redirect Dashboard
User /
Angular
Auth
Auth
User
JWT
Database

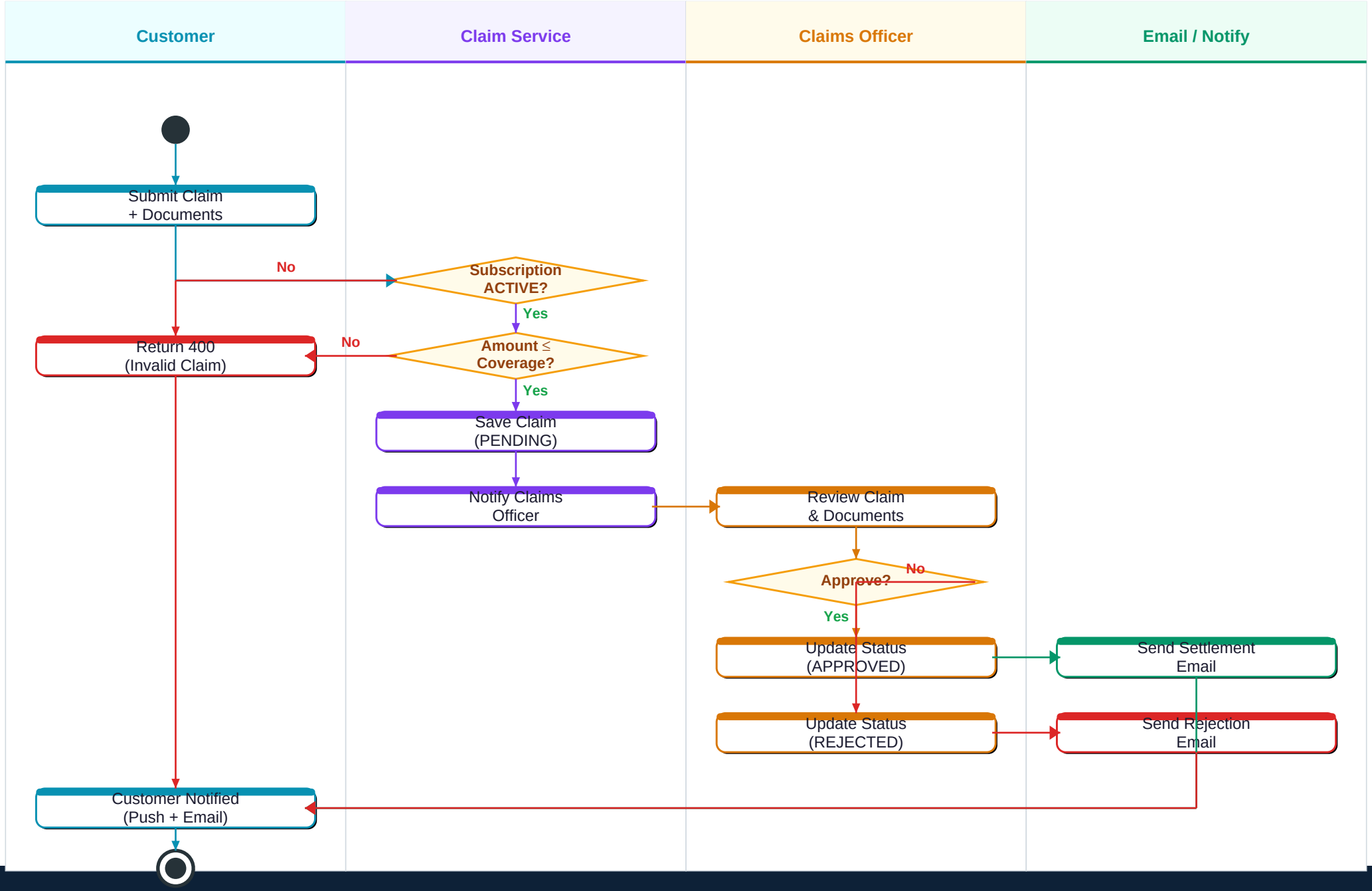

Customer
Angular Frontend
PolicyOrder Controller
PolicyOrder Service
Notification Service
Underwriter Portal
Subscription Repo
1 Select policy + vehicle + docs
2 POST /api/policy-orders
3 createOrder(userId, policyId)
4 save(PolicyOrder{PENDING})
5 orderId returned
6 notifyUnderwriter(orderId)
7 Push: new app pending review
8 201 Created {orderId, PENDING}
9 Show confirmation screen
10 Underwriter approves order
11 PUT /policy-orders/{id}/approve
12 validateDocuments() – gate check
13 save(PolicySubscription{ACTIVE})
14 Subscription created
15 notifyCustomer(activated)
16 200 OK – Subscription details
17 Dashboard – Policy ACTIVE
Customer
Angular
PolicyOrder
PolicyOrder
Notification
Underwriter
Subscription

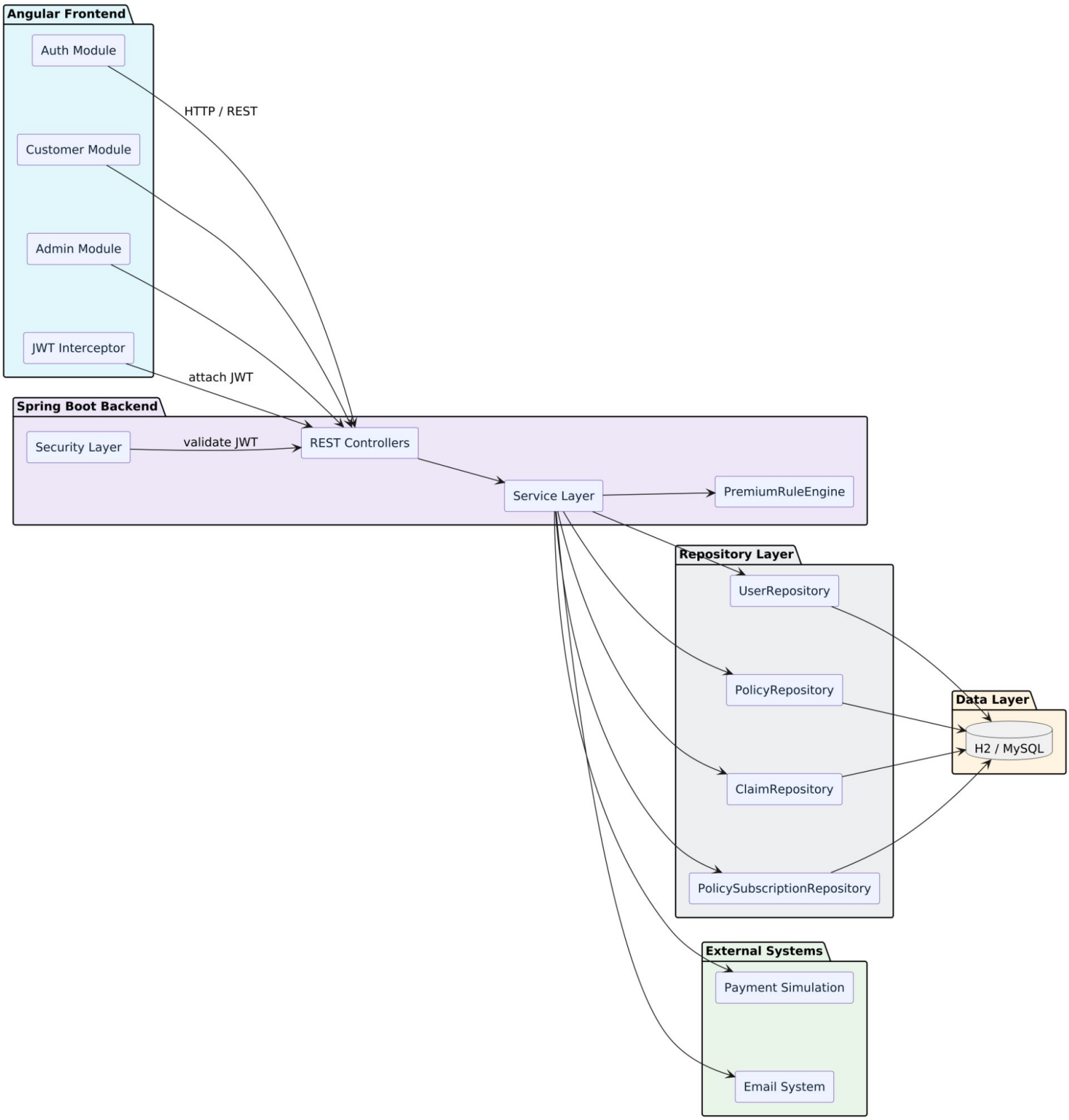

Customer
Angular Frontend
Premium Controller
Premium Service
PremiumRule Engine
UsageData Repo
PremCalc Repo
1 Open premium calculator
2 POST /api/premium/calculate/{subId}
3 calculatePremium(subId, usageId)
4 findByUsageId(usageId)
5 UsageData entity
6 applyRules(basePremium, usageData)
7 Rule: Distance >1000km → ×1.25
8 Rule: NightDriving >5h → ×1.10
9 Rule: HighRisk category → ×1.30
10 finalPremium = base × total multiplier
11 save(PremiumCalculation)
12 calculationId confirmed
13 200 OK {calculationId, finalPremium}
14 Display calculated premium
Customer
Angular
Premium
Premium
PremiumRule
UsageData
PremCalc

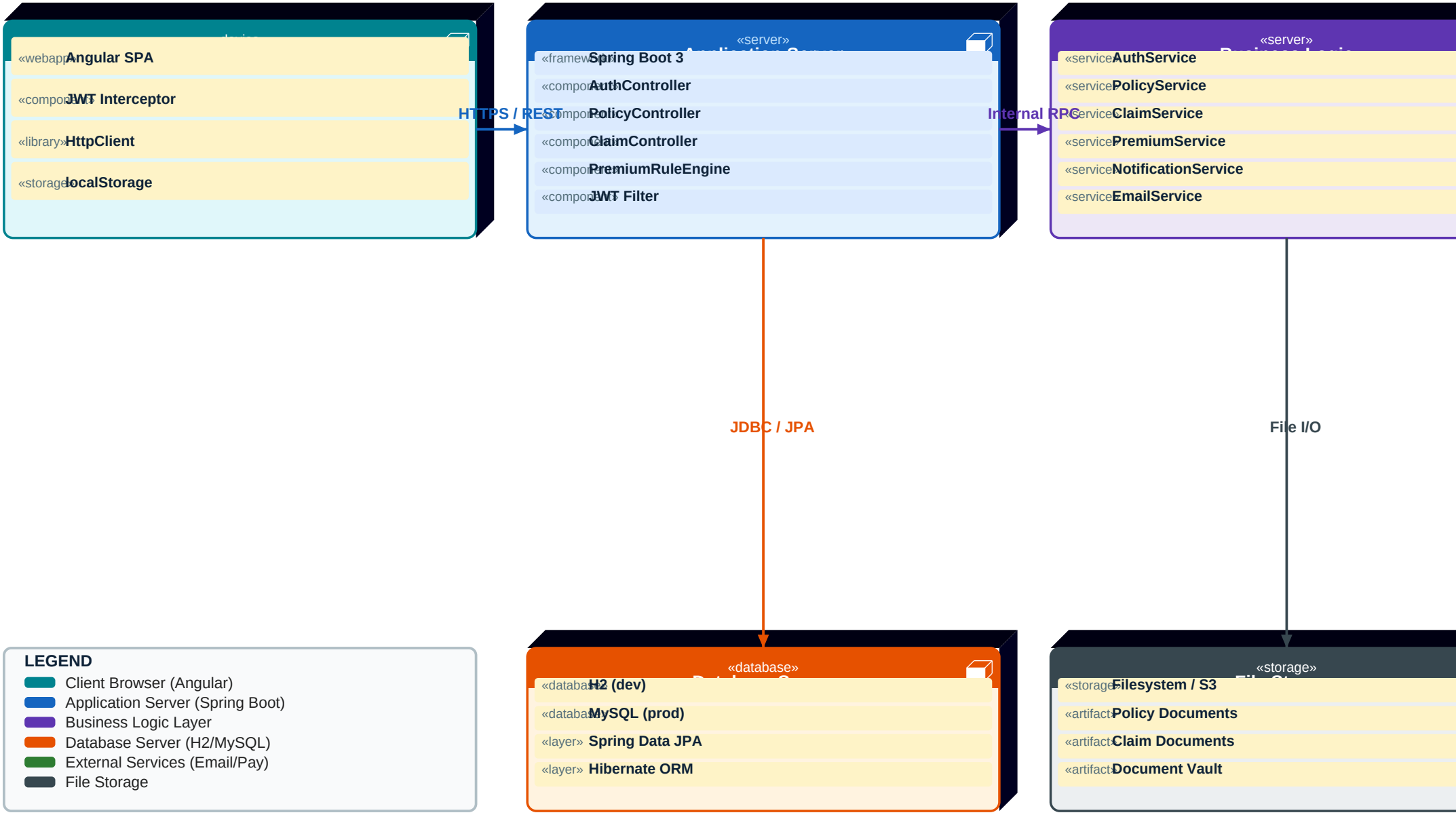

Customer
Angular Frontend
Claim Controller
Claim Service
Notification Svc
Claims Officer
Email Service
1 Submit claim + attach documents
2 POST /api/claims (multipart)
3 createClaim(subId, amount)
4 validateSubscriptionActive()
5 validateCoverageBalance(amount)
6 notifyClaimsOfficer(claimId)
7 Push: new claim pending review
8 201 Created {claimId, PENDING}
9 Claim submitted confirmation
10 Claims Officer approves claim
11 PUT /api/claims/{id}/approve
12 updateClaimStatus(APPROVED)
13 notifyCustomer(claim approved)
14 sendEmail(settlement letter)
15 Email delivered OK
16 200 OK {claimId, APPROVED}
17 Status APPROVED on dashboard
Customer
Angular
Claim
Claim
Notification
Claims
Email


Customer
PolicyOrder Service
Underwriter
System / DB
Select Policy & Vehicle
Upload KYC Documents
Create PolicyOrder (PENDING)
No
Documents Complete?
Yes
Notify Underwriter
Review Application & Risk
Approve?
Yes
No
Approve Order (APPROVED)
Reject Order (REJECTED)
Create Subscription (ACTIVE)
Notify Customer (Email+Push)


Customer / Telematics
Premium Service & Rule Engine
System / Database
Submit UsageData (km, night hrs)
Save UsageData to DB
PremiumService Triggers Calculation
Distance > 1000 km?
No
Yes
Apply ×1.25 (Distance Rule)
Night Driving > 5 hrs?
No
Yes
Apply ×1.10 (Night Rule)
Risk Category High?
No
Yes
Apply ×1.30 (High Risk Rule)
finalPremium = base × multipliers
Display Premium to Customer
Save PremiumCalc Record


Customer
Claim Service
Claims Officer
Email / Notify
Submit Claim + Documents
No
Subscription ACTIVE?
Yes
Return 400 (Invalid Claim)
No
Amount ≤ Coverage?
Yes
Save Claim (PENDING)
Notify Claims Officer
Review Claim & Documents
Approve?
No
Yes
Update Status (APPROVED)
Send Settlement Email
Update Status (REJECTED)
Send Rejection Email
Customer Notified (Push + Email)


Angular Frontend
Auth Module
Customer Module
Admin Module
JWT Interceptor
HTTP / REST
attach JWT
Spring Boot Backend
Security Layer
validate JWT
REST Controllers
Service Layer
PremiumRuleEngine
Repository Layer
UserRepository
PolicyRepository
ClaimRepository
PolicySubscriptionRepository
Data Layer
H2 / MySQL
External Systems
Payment Simulation
Email System


«webapp» Angular SPA
«component» JWT Interceptor
«library» HttpClient
«storage» localStorage
HTTPS / REST
«server»
«framework» Spring Boot 3
«component» AuthController
«component» PolicyController
«component» ClaimController
«component» PremiumRuleEngine
«component» JWT Filter
Internal RPC
«server»
«service» AuthService
«service» PolicyService
«service» ClaimService
«service» PremiumService
«service» NotificationService
«service» EmailService
JDBC / JPA
File I/O
«database»
«database» H2 (dev)
«database» MySQL (prod)
«layer» Spring Data JPA
«layer» Hibernate ORM
«storage»
«storage» Filesystem / S3
«artifact» Policy Documents
«artifact» Claim Documents
«artifact» Document Vault
LEGEND
Client Browser (Angular)
Application Server (Spring Boot)
Business Logic Layer
Database Server (H2/MySQL)
External Services (Email/Pay)
File Storage